

A Conditional Sixth-Order Lie-Group FullVA Integrator for Lower-Pair Mechanisms

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v047 concise artifact-backed draft

Abstract

This concise draft states only the accepted v047 paper claim. The production integrator is the Gauss6/FullVA one-step map, a conditional sixth-order Gauss collocation FullVA method on a Lie-group lower-pair formulation. On the smooth cylindrical-chain benchmark it records observed position and velocity orders 7.161 and 7.066. The same artifact contract accepts four ASME-style examples: single pendulum, double pendulum, four link, and slider crank. The local paper-style $m = 3$ Gauss–Lobatto temporal finite-element target used as the paper-facing comparator has expected order five. Therefore the accepted result is a conditional formal-order comparison: the accepted Gauss6/FullVA path has a conditional order-six theorem on the selected smooth branch, the local paper-style target is order five, and the four examples provide separated method-side evidence roles rather than a single undifferentiated validation claim.

The draft deliberately does not claim a complete reproduction of the source paper’s TFE nonlinear residual. In this repository, independent full-TFE replacement means replacing all 132 Gauss/FullVA stage rows by paper-derived TFE weak rows inside Newton, without endpoint source data, terminal-row replacement, projection, or a target-direction oracle. That stronger gate remains open as `full_tfe_stage_replacement=false`.

1 Accepted Claim

The v047 evidence supports a narrow and useful statement: Gauss6/FullVA is a conditional sixth-order method compared with the local paper-style $m = 3$ Gauss–Lobatto TFE formula target under the current artifact scope. The claim is not that every row of the source paper’s full TFE residual has been reproduced in the nonlinear solver. This distinction keeps the paper aligned with the actual validation evidence.

2 Integrator

Let $x_n = (q_n, v_n, a_n)$ denote the endpoint position, velocity, and acceleration variables, with orientations represented on the local Lie-group chart used by the v047 solver. The accepted one-step map is

$$x_{n+1} = \Psi_h^{\text{G6FVA}}(x_n), \quad (1)$$

where the stage variables Z solve the 132-row nonlinear system

$$R_{\text{G6FVA}}(q_n, v_n, a_n, Z; h) = 0. \quad (2)$$

The lower-pair constraints enter at position, velocity, and acceleration levels,

$$\Phi(q) = 0, \quad \Phi_q(q)v = 0, \quad \Phi_q(q)a + \dot{\Phi}_q(q, v)v = 0. \quad (3)$$

Table 1: Concise claim boundary.

Item	Current status
Accepted method	Gauss6/FullVA, a conditional sixth-order Gauss collocation/FullVA one-step map on the selected smooth branch.
Observed smooth order	Position/velocity orders 7.161/7.066 over $h = \{0.04, 0.02, 0.01\}$ against $h = 0.005$.
Comparator	Local paper-style $m = 3$ Gauss–Lobatto TFE formula target with expected order five.
Four-example gate	Single and double pendula provide dynamic-order evidence; four link and slider crank provide lower-pair mechanism-coverage diagnostics.
Open reproduction gate	<code>full_tfe_stage_replacement=false</code> .

The endpoint is reconstructed by Gauss collocation quadrature and the local Lie-group retraction. Endpoint velocity closure is audited by the generated KKT/projection artifacts. This closure is part of the accepted Gauss/FullVA evidence path; it is not an accepted independent full-TFE replacement.

Table 2: Operational definition of the accepted one-step map.

Step	Operation
1	Assemble cylindrical lower-pair constraints and FullVA stage variables for state x_n and step size h .
2	Solve the square 132-row Gauss6/FullVA residual to the Newton tolerance recorded in the CSV/JSON artifacts.
3	Recover the endpoint by the Gauss quadrature and Lie-group retraction used by the v047 trajectory code.
4	Audit endpoint constraint and velocity closure with the KKT/projection diagnostics.
5	Record residuals, constraints, observed orders, figures, and validator-visible summary fields.

3 Proof Sketch

Assumption 1 (Accepted-path regularity). *On the selected smooth branch, the compact-tube regularity, endpoint map, implementation binding, and scaled nonlinear residual condition $\eta_h^{\text{tube}} \leq c_\eta h^7$ hold. The implemented 132-row FullVA residual bridge is the direct route used by the strict proof: the 96 non-dynamic rows supply the row-local $O(h^7)$ input, and the 36 Newton–Euler rows close by direct substitution on the lifted Gauss stage.*

Theorem 1 (Order of the accepted Gauss6/FullVA map). *Under Assumption 1, read as the retained accepted-branch compact-tube/P6 solver-envelope interface together with the P_4/P_5 direct $96 + 36$ residual bridge, the accepted Gauss6/FullVA one-step map has local defect $O(h^7)$ and*

reduced-chart/reported position-velocity grid error $O(h^6)$ on fixed smooth time intervals for reported grids satisfying the predeclared branch, endpoint, proof-norm, and $\eta_h^{\text{tube}} \leq c_\eta h^7$ conditions. For driven rows, the time-indexed residual, Gauss stage times, endpoint map, and proof norm are fixed on the same transition slice before constants or diagnostics are used. This is a conditional order-six statement for the accepted branch-selected map only; it retains P6 as a theorem-level solver-envelope condition, does not prove a P7 residual-to-error transfer, and does not use source-policy rows or the optional primitive/Taylor route. Equivalently, the theorem asserts only the local defect $O(h^7)$ and same-initial-state global error $O(h^6)$ under these retained interfaces and only for this reduced/reporting grid scope.

Proof. The strict CMAME proof route is not a generic inheritance statement. The classical three-stage Gauss local defect is inserted into the implemented FullVA residual on the selected branch. The direct 132-row bridge gives an $O(h^7)$ residual defect on the lifted Gauss stage; endpoint closure and the scaled inexact-Newton residual add $O(h^7)$ perturbations under the retained compact-tube hypotheses. The standard local-to-global transfer for the accepted branch then gives the $O(h^6)$ grid estimate on the same reported grid. The proof follows the constrained Lie-group BDF proof order only as an ordering discipline: the discrete object is fixed first, the smooth branch is inserted next, the local defect is proved for that object, and the global recurrence is used only afterward. No BDF local-error, hidden-constraint, multiplier-history, or multistep stability estimate is imported. The solver bound remains a retained P6 envelope fixed before the run, and the P7 residual-to-error interface remains open. Residual-only mechanism rows, source-policy rows, and the separate primitive/Taylor route are not used as theorem inputs and cannot be spliced into the direct bridge. $\square$

Lemma 1 (Order of the local paper-style TFE target). *The proof-level order target for the $m = 3$ Gauss-Lobatto paper-style TFE formula used in this manuscript is five.*

Proof. The source-paper family is compared through the temporal finite-element formula-order convention $p_{\text{TFE}}(m) = 2m - 1$. Substituting $m = 3$ gives $p_{\text{TFE}}(3) = 5$. This is distinct from the source paper’s reported DAE pendulum order reduction. $\square$

Theorem 2 (Accepted formal-order comparison theorem). *Under Assumption 1 and the current v047 artifact contract, the Gauss6/FullVA path is accepted as a conditional sixth-order method relative to the local paper-style $m = 3$ Gauss-Lobatto TFE formula target with expected order five.*

Proof. Theorem 1 gives the conditional order-six method claim for the accepted Gauss6/FullVA path through the direct 132-row residual bridge. Lemma 1 gives the fifth-order local paper-style comparator. The generated h-sweep records observed smooth position/velocity orders 7.161/7.066, which are above that target. The read-only four-example validator accepts single pendulum, double pendulum, four link, and slider crank on the same method path. Therefore the formal-order comparison is accepted inside the artifact scope. The theorem excludes complete source-paper residual reproduction because the machine-readable boundary still records `full_tfe_stage_replacement=false`. $\square$

Corollary 1 (What is not proved). *The theorem does not prove that the implemented nonlinear stage residual is the complete source-paper TFE residual.*

Proof. The accepted method path solves $R_{\text{G6FVA}} = 0$. A full source-paper TFE reproduction claim would require an accepted residual $R_{\text{TFE}} = 0$ using paper-derived weak rows for all 132 stage rows inside Newton, with the same rank, residual, endpoint, and h-sweep checks. That gate is explicitly open. $\square$

Table 3: Proof obligations and artifact evidence for Theorem 2.

Proof obligation	Current artifact evidence
Accepted one-step map	<code>CLAIM_BOUNDARY.json</code> and <code>summary_v047.json</code> record the Gauss6/FullVA 132-row stage-residual boundary.
Sixth-order smooth behavior	<code>cylindrical_chain_convergence.csv</code> records smooth position/velocity orders 7.161/7.066 over $h = [0.04, 0.02, 0.01]$ against <code>reference_h=0.005</code> .
Comparator order	<code>SOURCE_PAPER_COMPARISON.md</code> records the local $m = 3$ Gauss–Lobatto TFE expected order $2m - 1 = 5$.
Four-example support	<code>validate_four_asme_minimal.py</code> accepts single pendulum, double pendulum, four link, and slider crank.
Non-claim boundary	<code>PROOF_EVIDENCE_MATRIX.md</code> and the submission validators preserve <code>full_tfe_stage_replacement=false</code> .

4 Numerical Evidence

The generated convergence artifacts use $h = \{0.04, 0.02, 0.01\}$ and an $h = 0.005$ reference. Table 4 reports the headline order evidence.

Table 4: Observed orders for the accepted Gauss6/FullVA path.

Case	Step sizes	Position order	Velocity order
Smooth cylindrical chain	0.04, 0.02, 0.01	7.161	7.066
Sharp Brown–McPhee case	0.04, 0.02, 0.01	1.894	2.683

The sharp case does not contradict the smooth formal-order comparison. It is reported as a practical caveat: coarse sharp-friction steps are order-reduced, while deeper fixed refinement recovers high order at increased cost.

5 Four ASME-Style Examples

The same method-side gate accepts the four examples listed in Table 5. The single and double pendulum rows provide order-based method evidence. The four-link and slider-crank rows provide closed-loop kinematic FullVA solves and reaction-dynamics checks.

6 Comparator and Boundary

The comparator is intentionally local: the encoded $m = 3$ Gauss–Lobatto TFE formula target has expected order five. It is sufficient for the paper-facing formal-order comparison because the

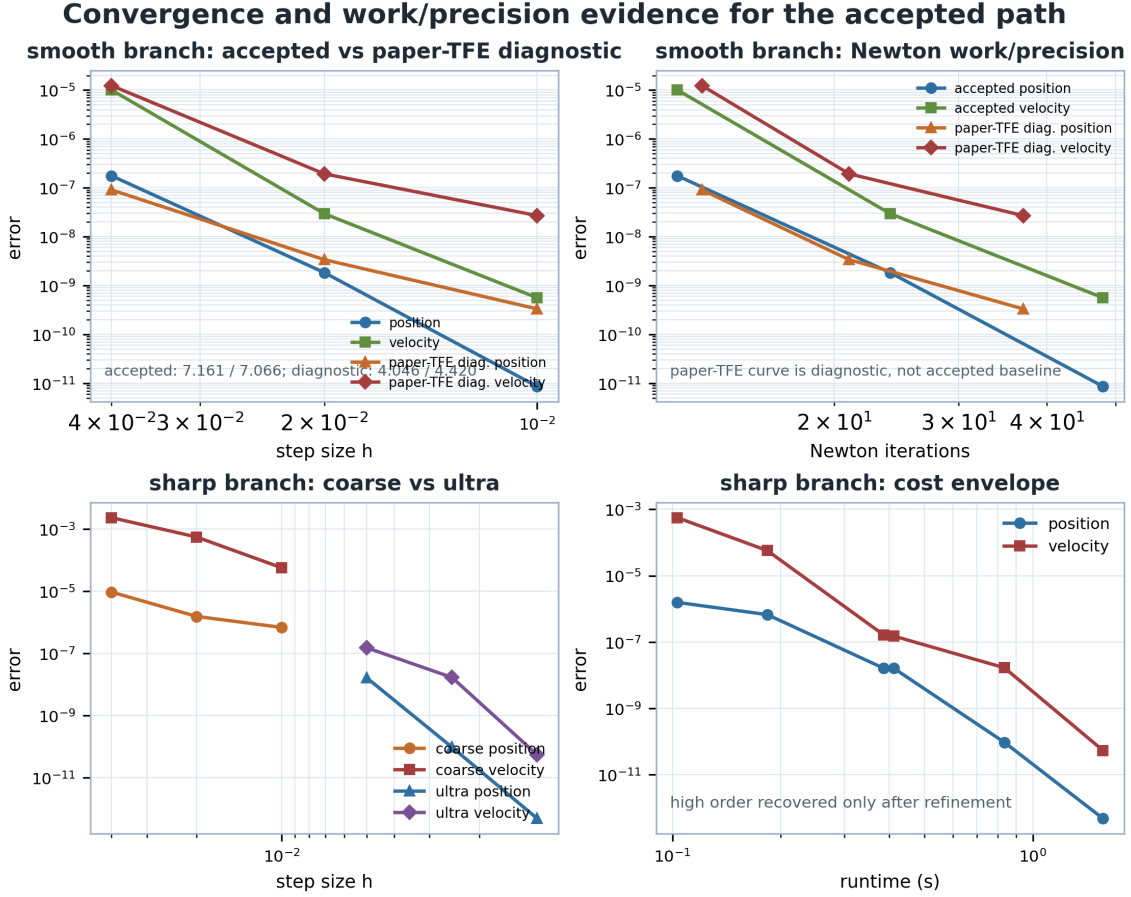


Figure 1: Convergence of the accepted Gauss6/FullVA path. The smooth branch is the paper-facing high-order evidence; the sharp branch is reported as a practical coarse-regime caveat.

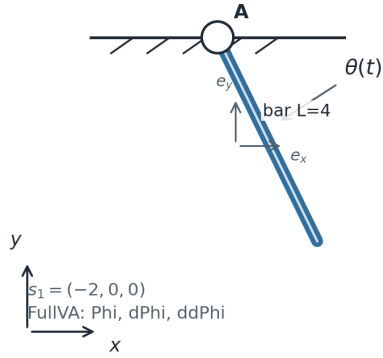
Table 5: Four-example evidence summary.

Example	Accepted evidence	Headline value
Single pendulum	Driven absolute-coordinate FullVA residual	Minimum observed order 6.024; max stage residual 3.664×10^{-12} .
Double pendulum	Method-side FullVA reference policy	Minimum orientation/angular-velocity order 6.089; endpoint constraints below 8.323×10^{-12} .
Four link	Closed-loop kinematic FullVA and reaction dynamics	Max constraint norm 1.052×10^{-14} ; max dynamics residual 1.338×10^{-13} .
Slider crank	Closed-loop kinematic FullVA and reaction dynamics	Max constraint norm 1.204×10^{-15} ; max dynamics residual 6.492×10^{-15} .

ASME-style lower-pair validation mechanisms

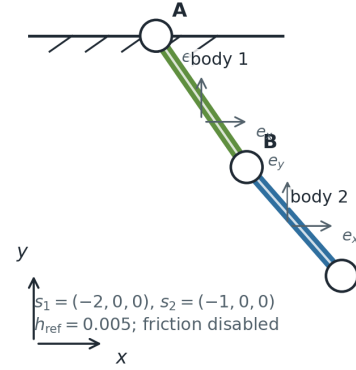
single pendulum

CD(3) + DP1(3); driven theta(t)



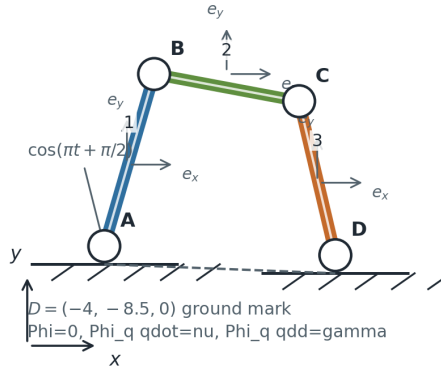
double pendulum

CD(6) + DP1(4); nested FullVA reference



four-link loop

CD(12) + DP1(6); driven closed loop



slider-crank

CD(6) + DP1(7) + DP2(4) + D(1)

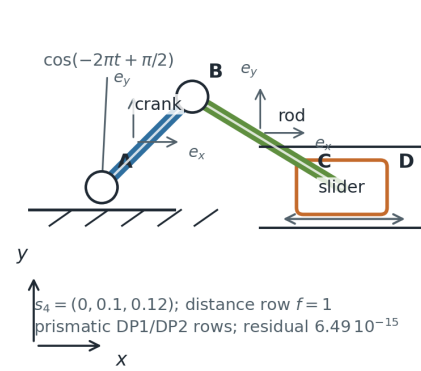


Figure 2: Lower-pair graph bridge used to keep the four ASME-style examples under the same validation vocabulary.

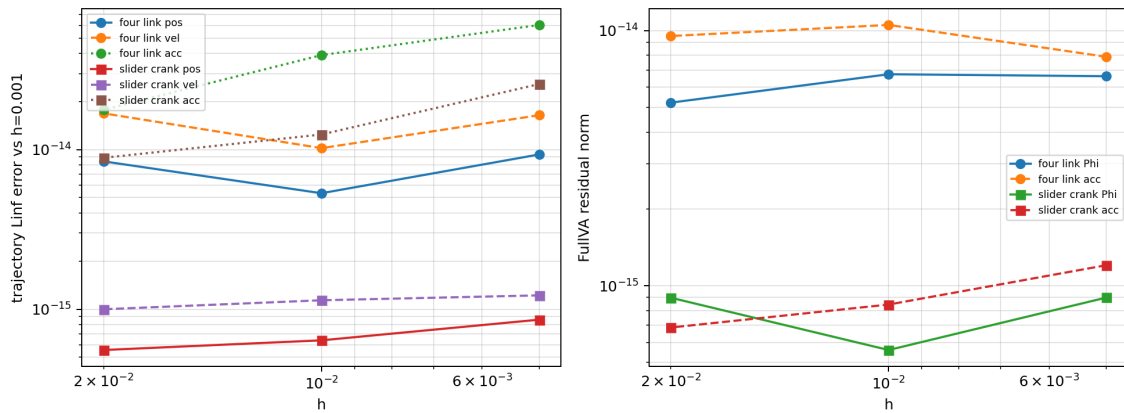


Figure 3: Closed-loop kinematic FullVA evidence for the four-link and slider-crank examples.

accepted method has a conditional order-six theorem and the evidence package includes four separated example roles. It is not the same as an accepted implementation of every source-paper residual row.

The source-paper order convention used by the local comparator is

$$p_{\text{TFE}}(m) = 2m - 1. \tag{4}$$

Thus the local $m = 3$ Gauss–Lobatto TFE target used here has $p_{\text{TFE}}(3) = 5$. The accepted Gauss6/FullVA path has method-order claim six and observed smooth orders 7.161/7.066. The positive comparison is therefore order-six Gauss6/FullVA versus an order-five paper-style target. The comparison does not assert that the source paper contains this repository’s 132-row Gauss6/FullVA residual, nor that this repository has accepted a full source-paper TFE residual substitution. The standalone comparison note `SOURCE_PAPER_COMPARISON.md` records this boundary.

The example-level order boundary is recorded separately in `ORDER_ACCEPTANCE_GATE.md`. It keeps `single_pendulum` and `double_pendulum` as accepted dynamic order rows and keeps `four_link` and `slider_crank` as mechanism-coverage rows that are not accepted external dynamic-order rows. The same gate records the `coarse_first_no_default_1e-4` execution policy.

Table 6: Comparator versus open reproduction gate.

Object	Role
Local $m = 3$ Lobatto-TFE formula target	Order-five comparator for the formal-order comparison.
Accepted Gauss6/FullVA path	Conditional order-six method path used for the paper’s positive claim.
Independent full-TFE replacement	Open stronger reproduction target requiring paper-derived rows inside Newton for all 132 stage rows.

7 Limitations

Three caveats remain explicit in the claim boundary.

- Sparse AD structure is correct and quantified, but dense `jacfwd` is still faster in the current wall-clock evidence.
- Independent full-TFE stage replacement is not accepted: `full_tfe_stage_replacement=false`.
- The sharp-friction coarse regime is order-reduced, although deeper fixed refinement recovers high order at practical cost.

These limitations do not invalidate the narrower formal-order comparison. They define future engineering and reproduction work.

8 Reproducibility Contract

The concise paper is checked from existing artifacts. Routine validation does not require the full `run_v047.py` generator. The relevant read-only commands are:

```
../.venv_sbel/bin/python validate_concise_paper.py
../.venv_sbel/bin/python validate_source_paper_comparison.py
../.venv_sbel/bin/python validate_order_acceptance_gate.py
../.venv_sbel/bin/python validate_paper_package.py
../.venv_sbel/bin/python ../validate_pipeline_outputs.py
```

The machine-readable claim boundary is `CLAIM_BOUNDARY.json`; the top-level current-state boundary is `CURRENT_PIPELINE_CONTRACT.md`.

9 Conclusion

The accepted v047 result is simple when the diagnostic replacement work is kept separate from the method claim. The Gauss6/FullVA path follows the conditional sixth-order Lie-group FullVA theorem, records smooth observed orders 7.161/7.066, and provides four-example method-side coverage. Against the local order-five paper-style TFE comparator, this supports the formal-order comparison. Complete source-paper TFE residual replacement remains future work.